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FIELD EFFECT TRANSISTOR WITH SOURCE/DRAIN CONTACT ISOLATION STRUCTURE AND METHOD

Abstract

A device includes a substrate and a gate structure wrapping around at least one vertical stack of nanostructure channels. The device includes a source/drain region abutting the gate structure, and a source/drain contact over the source/drain region. The device includes an etch stop layer laterally between the source/drain contact and the gate structure and having a first sidewall in contact with the source/drain contact, and a second sidewall opposite the first sidewall. The device includes a source/drain contact isolation structure embedded in the source/drain contact and having a third sidewall substantially coplanar with the second sidewall of the etch stop layer.

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Background/Summary

BACKGROUND

[0001] The semiconductor integrated circuit (IC) industry has experienced exponential growth. Technological advances in IC materials and design have produced generations of ICs where each generation has smaller and more complex circuits than the previous generation. In the course of IC evolution, functional density (i.e., the number of interconnected devices per chip area) has generally increased while geometry size (i.e., the smallest component (or line) that can be created using a fabrication process) has decreased. This scaling down process generally provides benefits by increasing production efficiency and lowering associated costs. Such scaling down has also increased the complexity of processing and manufacturing ICs.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0002] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0003] FIGS. **1A-1F** are diagrammatic top and cross-sectional side views of a portion of an IC device fabricated according to embodiments of the present disclosure.

[0004] FIGS. **2A-16C** are views of various embodiments of an IC device of at various stages of fabrication according to various aspects of the present disclosure.

[0005] FIG. **17** is a flowchart illustrating a method of fabricating a semiconductor device according to various aspects of the present disclosure.

DETAILED DESCRIPTION

[0006] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0007] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship

to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0008] Terms indicative of relative degree, such as “about,” “substantially,” and the like, should be interpreted as one having ordinary skill in the art would in view of current technological norms. Generally, the term “substantially” indicates a tighter tolerance than the term “about.” For example, a thickness of “about 100 units” will include a larger range of values, e.g., 70 units to 130 units (+/-30%), than a thickness of “substantially 100 units,” which will include a smaller range of values, e.g., 95 units to 105 units (+/-5%). Again, such tolerances (+/-30%, +/-5%, and the like) may be process- and/or equipment-dependent, and should not be interpreted as more or less limiting than a person having ordinary skill in the art would recognize as normal for the technology under discussion, other than that “about” as a relative term is not as stringent as “substantially” when used in a similar context.

[0009] The present disclosure is generally related to semiconductor devices, and more particularly to field-effect transistors (FETs), such as planar FETs, three-dimensional fin-line FETs (FinFETs), or gate-all-around (GAA) devices. In advanced technology nodes, dimension scaling can lead to difficulties forming isolated contacts to features of the semiconductor devices. Source/drain isolation structures formed by methods described herein achieve improved yield by reducing sidewall merging that leads to voids in the isolation structures. The merging is reduced or eliminated by removing (partially or fully) a contact etch stop layer (CESL) from sidewalls prior to depositing material of the isolation structures. Removal of the CESL prior to deposition increases spacing between the sidewalls, which reduces the occurrence of merging during deposition.

[0010] The gate all around (GAA) transistor structures may be patterned by any suitable method. For example, the structures may be patterned using one or more photolithography processes, including double-patterning or multi-patterning processes. Generally, double-patterning or multi-patterning processes combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. For example, in one embodiment, a sacrificial layer is formed over a substrate and patterned using a photolithography process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers may then be used to pattern the GAA structure.

[0011] FIGS. 1A-1F illustrate diagrammatic top and cross-sectional side views of a portion of an IC device **10** fabricated according to embodiments of the present disclosure, where the IC device **10** includes gate-all-around (GAA) devices **20A-20E**. FIG. 1A is a diagrammatic top view of the portion of the IC device **10** including the GAA devices **20A-20E**. FIG. 1B is a cross-sectional side view of a portion of the IC device **10** along the line B-B shown in FIG. 1A. FIG. 1C is a cross-sectional side view of single GAA device **20B**. FIG. 1D is a cross-sectional side view of a portion of the IC device **10** along the line D-D shown in FIG. 1A. FIG. 1E is a cross-sectional side view of a portion of the IC device **10** along the line E-E shown in FIG. 1A. FIG. 1F is a cross-sectional side view of a portion of the IC device **10** along the line E-E shown in FIG. 1A in accordance with various embodiments. Certain features are removed from view intentionally in the top view of FIG. 1A for simplicity of illustration.

[0012] The GAA devices **20A-20E** may include at least an N-type FET (NFET) or a P-type FET (PFET) in some embodiments. Integrated circuit devices such as the IC device **10** frequently include transistors having different threshold voltages based on their function in the IC device. For example, input/output (IO) transistors typically have the highest threshold voltages, core logic transistors typically have the lowest threshold voltages, and a third threshold voltage between that of the IO transistors and that of the core logic transistors may also be employed for certain other

functional transistors, such as static random access memory (SRAM) transistors. Some circuit blocks within the IC device **10** may include two or more NFETs and/or PFETs of two or more different threshold voltages.

[0013] Referring to FIGS. **1A-1E**, the GAA devices **20A-20E** (see FIG. **1D**) are formed over and/or in a substrate **110**, and generally include gate structures **200A-200E** straddling and/or wrapping around semiconductor channels **22A1-22B5**, alternately referred to as “nanostructures,” located over semiconductor fins **321-325** protruding from, and separated by, isolation structures **361-364** (see FIG. **1E**). The channels are labeled “**22AX**” to “**22BX**,” where “**X**” is an integer from 1 to 5, corresponding to the five transistors **20A-20E**, respectively. Each gate structure **200A-200E** controls current flow through the channels **22A1-22B5**.

[0014] Referring to FIG. **1C**, the cross-sectional view of the IC device **10** in FIG. **1D** is taken along an X-Z plane, where the X-direction is the horizontal direction, and the Z-direction is the vertical direction. The cross-sectional view in FIG. **1C** shows a single GAA device **20B** of the GAA devices **20A-20E** for simplicity of illustration, and the related description is generally applicable to the other GAA devices **20A**, **20C-20E**. In FIG. **1C**, the GAA device **20B** is shown including three channels **22A2-22C2**, which are laterally abutted by source/drain features **82**, and covered and surrounded by the gate structure **200B**. Generally, the number of channels **22** is two (as shown in FIG. **1D**) or more than two, such as three (FIG. **1C**) or four or more. The gate structure **200B** controls flow of electrical current through the channels **22A2-22C2** to and from the source/drain features **82** based on voltages applied at the gate structure **200B** and at the source/drain features **82**.

[0015] In some embodiments, the fin structure **322** includes silicon. In some embodiments, the GAA device **20B** is an NFET, and the source/drain features **82** thereof include silicon phosphorous (SiP). In some embodiments, the GAA device **20B** is a PFET, and the source/drain features **82** thereof include silicon germanium (SiGe).

[0016] The channels **22A2-22C2** each include a semiconductive material, for example silicon or a silicon compound, such as silicon germanium, or the like. The channels **22A2-22C2** are nanostructures (e.g., having sizes that are in a range of a few nanometers) and may also each have an elongated shape and extend in the X-direction. In some embodiments, the channels **22A2-22C2** each have a nano-wire (NW) shape, a nano-sheet (NS) shape, a nano-tube (NT) shape, or other suitable nanoscale shape. The cross-sectional profile of the channels **22A2-22C2** may be rectangular, round, square, circular, elliptical, hexagonal, or combinations thereof.

[0017] In some embodiments, the lengths (e.g., measured in the X-direction) of the channels **22A2-22C2** may be different from each other, for example due to tapering during a fin etching process. In some embodiments, length of the channel **22A1** may be less than a length of the channel **22B1**, which may be less than a length of the channel **22C1**. The channels **22A2-22C2** each may not have uniform thickness, for example due to a channel trimming process used to expand spacing (e.g., measured in the Z-direction) between the channels **22A2-22C2** to increase gate structure fabrication process window. For example, a middle portion of each of the channels **22A2-22C2** may be thinner than the two ends of each of the channels **22A2-22C2**. Such shape may be collectively referred to as a “dog-bone” shape.

[0018] In some embodiments, the spacing between the channels **22A2-22C2** (e.g., between the channel **22B2** and the channel **22A2** or the channel **22C2**) is in a range between about 8 nanometers (nm) and about 12 nm. In some embodiments, a thickness (e.g., measured in the Z-direction) of each of the channels **22A2-22C2** is in a range between about 5 nm and about 8 nm. In some embodiments, a width (e.g., measured in the Y-direction, not shown in FIG. **1C**, orthogonal to the X-Z plane) of each of the channels **22A2-22C2** is at least about 8 nm.

[0019] The gate structure **200B** is disposed over and between the channels **22A2-22C2**, respectively. In some embodiments, the gate structure **200B** is disposed over and between the channels **22A2-22C2**, which are silicon channels for N-type devices or silicon germanium channels for P-type devices. In some embodiments, the gate structure **200B** includes an interfacial layer (IL)

210, one or more gate dielectric layers **600**, one or more work function tuning layers **900**, and a metal fill layer **290**.

[0020] The interfacial layer **210**, which may be an oxide of the material of the channels **22A2-22C2**, is formed on exposed areas of the channels **22A2-22C2** and the top surface of the fin **322**. The interfacial layer **210** promotes adhesion of the gate dielectric layers **600** to the channels **22A2-22C2**. In some embodiments, the interfacial layer **210** has thickness of about 5 Angstroms (Å) to about 50 Angstroms (Å). In some embodiments, the interfacial layer **210** has thickness of about 10 Å. The interfacial layer **210** having thickness that is too thin may exhibit voids or insufficient adhesion properties. The interfacial layer **210** being too thick consumes gate fill window, which is related to threshold voltage tuning and resistance as described above. In some embodiments, the interfacial layer **210** is doped with a dipole, such as lanthanum, for threshold voltage tuning.

[0021] In some embodiments, the gate dielectric layer **600** includes at least one high-k gate dielectric material, which may refer to dielectric materials having a high dielectric constant that is greater than a dielectric constant of silicon oxide ($k \approx 3.9$). Exemplary high-k dielectric materials include HfO_2 , HfSiO , HfSiON , HfTaO , HfTiO , HfZrO , ZrO_2 , Ta_2O_5 , or combinations thereof. In some embodiments, the gate dielectric layer **600** has thickness of about 5 Å to about 100 Å.

[0022] In some embodiments, the gate dielectric layer **600** may include dopants, such as metal ions driven into the high-k gate dielectric from La_2O_3 , MgO , Y_2O_3 , TiO_2 , Al_2O_3 , Nb_2O_5 , or the like, or boron ions driven in from B_2O_3 , at a concentration to achieve threshold voltage tuning. As one example, for N-type transistor devices, lanthanum ions in higher concentration reduce the threshold voltage relative to layers with lower concentration or devoid of lanthanum ions, while the reverse is true for P-type devices. In some embodiments, the gate dielectric layer **600** of certain transistor devices (e.g., IO transistors) is devoid of the dopant that is present in certain other transistor devices (e.g., N-type core logic transistors or P-type IO transistors). In N-type IO transistors, for example, relatively high threshold voltage is desirable, such that it may be preferable for the IO transistor high-k dielectric layers to be free of lanthanum ions, which would otherwise reduce the threshold voltage.

[0023] In some embodiments, the gate structure **200B** further includes one or more work function metal layers, represented collectively as work function metal layer **900**. When configured as an NFET, the work function metal layer **900** of the GAA device **20B** may include at least an N-type work function metal layer, an in-situ capping layer, and an oxygen blocking layer. In some embodiments, the N-type work function metal layer is or comprises an N-type metal material, such as TiAlC , TiAl , TaAlC , TaAl , or the like. The in-situ capping layer is formed on the N-type work function metal layer, and may comprise TiN , TiSiN , TaN , or another suitable material. The oxygen blocking layer is formed on the in-situ capping layer to prevent oxygen diffusion into the N-type work function metal layer, which would cause an undesirable shift in the threshold voltage. The oxygen blocking layer may be formed of a dielectric material that can stop oxygen from penetrating to the N-type work function metal layer, and may protect the N-type work function metal layer from further oxidation. The oxygen blocking layer may include an oxide of silicon, germanium, SiGe , or another suitable material. In some embodiments, the work function metal layer **900** includes more or fewer layers than those described.

[0024] The work function metal layer **900** may further include one or more barrier layers comprising a metal nitride, such as TiN , WN , MON , TaN , or the like. Each of the one or more barrier layers may have thickness ranging from about 5 Å to about 20 Å. Inclusion of the one or more barrier layers provides additional threshold voltage tuning flexibility. In general, each additional barrier layer increases the threshold voltage. As such, for an NFET, a higher threshold voltage device (e.g., an IO transistor device) may have at least one or more than two additional barrier layers, whereas a lower threshold voltage device (e.g., a core logic transistor device) may have few or no additional barrier layers. For a PFET, a higher threshold voltage device (e.g., an IO

transistor device) may have few or no additional barrier layers, whereas a lower threshold voltage device (e.g., a core logic transistor device) may have at least one or more than two additional barrier layers. In the immediately preceding discussion, threshold voltage is described in terms of magnitude. As an example, an NFET IO transistor and a PFET IO transistor may have similar threshold voltage in terms of magnitude, but opposite polarity, such as +1 Volt for the NFET IO transistor and -1 Volt for the PFET IO transistor. As such, because each additional barrier layer increases threshold voltage in absolute terms (e.g., +0.1 Volts/layer), such an increase confers an increase to NFET transistor threshold voltage (magnitude) and a decrease to PFET transistor threshold voltage (magnitude).

[0025] The gate structure **200B** also includes metal fill layer **290**. The metal fill layer **290** may include a conductive material such as tungsten, cobalt, ruthenium, iridium, molybdenum, copper, aluminum, or combinations thereof. Between the channels **22A2-22C2**, the metal fill layer **290** is circumferentially surrounded (in the cross-sectional view) by the one or more work function metal layers **900**, which are then circumferentially surrounded by the gate dielectric layers **600**. The gate structure **200B** may also include a glue layer that is formed between the one or more work function layers **900** and the metal fill layer **290** to increase adhesion. The glue layer is not specifically illustrated in FIGS. **1A-1F** for simplicity.

[0026] The GAA devices **20A-20E** also include gate spacers **41** and inner spacers **74** that are disposed on sidewalls of the gate dielectric layer **600** and the IL **210**. The inner spacers **74** are also disposed between the channels **22A2-22C2**. The gate spacers **41** and the inner spacers **74** may include a dielectric material, for example a low-k material such as SiOCN, SiON, SiN, or SiOC. In some embodiments, one or more additional spacer layers **49** are present abutting the gate spacers **41**, as shown in FIG. **1C**.

[0027] The GAA devices **20A-20E** may further include source/drain contacts **120M**, **120S** (shown in FIG. **1A**; collectively referred to as “source drain contacts **120**”) that are formed over the source/drain features **82**. The source/drain contacts **120** may include a conductive material such as tungsten, ruthenium, cobalt, copper, titanium, titanium nitride, tantalum, tantalum nitride, iridium, molybdenum, nickel, aluminum, or combinations thereof. The source/drain contacts **120** may be surrounded by barrier layers (not shown), such as SiN or TiN, which help prevent or reduce diffusion of materials from and into the source/drain contacts **120**. A silicide layer **118** may also be formed between the source/drain features **82** and the source/drain contacts **120**, so as to reduce the source/drain contact resistance. The silicide layer **118** includes nickel, cobalt, titanium, tantalum, platinum, tungsten, other noble metals, other refractory metals, rare earth metals or their alloys. In some embodiments, thickness of the silicide layer **118** (in the Z direction) is in a range of about 0.5 nm to about 5 nm. In some embodiments, height of the source/drain contacts **120** may be in a range of about 1 nm to about 50 nm.

[0028] As shown in FIGS. **1A** and **1D-1F**, one or more of the source/drain contacts, labeled **120M**, may each overlie multiple GAA devices along the Y-direction. In some embodiments, others of the source/drain contacts, labeled **120S**, may each overlie a single GAA device along the Y-direction. FIG. **1E** illustrates a cross-sectional side view in the Y-Z plane along the line E-E of FIG. **1A**, in which source/drain contacts **120S** overlie source/drain regions **82** and are separated by source/drain contact isolation structures **150**. Formation of the source/drain contact isolation structures **150** is described in further detail with reference to FIGS. **13A-16C**.

[0029] In some embodiments, each of the source/drain regions **82** is formed over a respective fin **322**, **323**, **324**, and is separated from others of the source/drain regions **82** by hybrid fins **94** (or “inactive fins **94**”) formed over isolation regions **361-364**. In some embodiments, the isolation regions **361-364** are shallow trench isolation (“STI”) regions. In some embodiments, each of the hybrid fins **94** includes a liner layer **93** (or “dielectric layer **93**”) and a fill layer **95** (or “oxide layer **95**”). Each of the source/drain contact isolation structures **150** may extend from an upper surface of a respective hybrid fin **94** to at least upper surfaces of the source/drain contacts **120S** abutting

either side of the source/drain contact isolation structure **150**, such that the source/drain contact isolation structure **150** completely electrically isolates the source drain contact **120S** on a first side of the source/drain contact isolation structure **150** from the source drain contact **120S** on a second opposite side of the source/drain contact isolation structure **150**. In the Y direction, each of the source/drain contact isolation structures **150** may extend from a first sidewall of the source/drain contact **120S** on the first side of the source/drain contact isolation structure **150** to a second side wall of the source/drain contact **120S** on the second side of the source/drain contact isolation structure **150**. In some embodiments, each of the source/drain contact isolation structures **150** has a first sidewall (or first sidewall portion) in direct physical contact with the source/drain contact **120S** on its first side and a second side wall (or second side wall portion) in direct physical contact with the source/drain contact **120S** on its second side.

[0030] In some embodiments, each of the source/drain contact isolation structures **150** is or includes SiN, SiCN, SiC, SiOC, SiOCN, HfO₂, ZrO₂, ZrAlO_x, HfAlO_x, HfSiO_x, Al₂O₃, or another suitable dielectric material. In some embodiments, width of the source/drain contact isolation structures **150** (along the Y direction) is in a range of about 8 nm to about 30 nm. The source/drain contact isolation structures **150** are formed following thinning or complete removal of an etch stop layer **131**, as described in greater detail with reference to FIGS. **13A-16C**. As such, aspect ratio, which is a ratio of height of the source/drain contact isolation structures **150** (Z-axis direction) over width of the source/drain contact isolation structures **150** (X-axis direction), is less than 7, such as less than about 5. The aspect ratio being less than about 5 is generally associated with an intact source/drain contact isolation structure **150**, meaning the source/drain contact isolation structure **150** may be substantially free of voids that form by upper portions of the source/drain contact isolation structure **150** merging during deposition prior to middle or lower portions of the source/drain contact isolation structure **150** merging.

[0031] FIGS. **1D** and **1E** show a configuration in which the etch stop layer **131** is completely removed prior to forming the source/drain contact isolation structures **150**. In FIG. **1F**, a configuration is shown in which the etch stop layer, labeled “**131CMD**,” is thinned and not completely removed. In some embodiments, prior to forming the source/drain contact isolation structures **150**, the etch stop layer **131CMD** is formed by thinning the etch stop layer **131** originally present in the space in which the source/drain contact isolation structures **150** are to be formed. In some embodiments, following the thinning, the etch stop layer **131CMD** has a tapered profile, as shown in FIG. **1F**. Thickness of the etch stop layer **131CMD** may be in a range of about 0.1 nm to about 2 nm. In some embodiments, the tops of the etch stop layer **131CMD** may be recessed relative to the top of the source/drain contact isolation structures **150**, such that an upper portion of sidewalls of the source/drain contact isolation structures **150** are in direct physical contact with neighboring structures, such as self-aligned contact (“SAC”) gate protection structures **295** and/or the source/drain contacts **120**. Generally, thickness of the etch stop layer **131CMD** does not exceed thickness of the etch stop layer **131**. In some embodiments, ratio of the thickness of the etch stop layer **131** over the thickness of the etch stop layer **131CMD** is in a range of about 2 to about 10. The ratio being less than about 2 may lead to insufficient aspect ratio in the source/drain contact isolation structures **150**, such that voids may form during formation of the source/drain contact isolation structures **150**.

[0032] Again to FIG. **1D**, certain of the GAA devices, such as the GAA devices **20D**, **20E**, further include an interlayer dielectric (ILD) **130**. The ILD **130** provides electrical isolation between the various components of the GAA devices **20D**, **20E** discussed above, for example between the gate structures **200D**, **200E** and the source/drain contact **120M** therebetween. The etch stop layer **131** may be formed prior to forming the ILD **130**, and may be positioned laterally between the ILD **130** and the gate spacers **41** and vertically between the ILD **130** and the source/drain features **82**. In some embodiments, the etch stop layer **131** is or includes SiN, SiCN, SiC, SiOC, SiOCN, HfO₂, ZrO₂, ZrAlO_x, HfAlO_x, HfSiO_x, Al₂O₃, or other suitable

material. In some embodiments, thickness of the etch stop layer **131** is in a range of about 1 nm to about 5 nm.

[0033] Vias **183**, **184**, which include source/drain vias **183** and gate via **184** are shown in accordance with various embodiments. In some embodiments, an etch stop layer **185** overlies the source/drain contacts **120M/120S**, the capping layer **295**, and the source/drain contact isolation structures **150**. A dielectric layer **181**, such as a second ILD **181**, overlies the etch stop layer **185**. In some embodiments, the dielectric layer **181** is or includes a low-k dielectric material, such as SiO, SiOC, SiON, SiOCN, or the like, and the etch stop layer **185** is or includes a different dielectric material, such as SiN, or other suitable material. The gate via **184** extends through the dielectric layer **181**, the etch stop layer **185** and the capping layer **295**, and lands on the conductive layer **204** or the gate fill layer **290** of the gate structure **200C**. The source/drain vias **183** extend through the dielectric layer **181** and the etch stop layer **185**, and land on the source/drain contacts **120M**, as shown in FIG. **1B**. A conductive line **182**, which may include a wire or trace, overlies and contacts the gate via **184** and the source/drain vias **183**. The conductive line **182** may be or include a metal, such as copper, aluminum, tungsten, or other appropriate conductive material.

[0034] As shown in FIGS. **1B**, **1F**, the gate structures **200B-200E** may straddle the hybrid fin **94** to form interconnection between transistors on either side of the hybrid fin **94**. In some embodiments, one or more of the gate structures **200A-200E**, such as the gate structure **200A**, may be electrically isolated by a gate isolation structure **99**. In some embodiments, the gate isolation structure **99** is or includes one or more dielectric materials, such as a high-k liner layer and a low-k fill layer.

[0035] Additional details pertaining to the fabrication of GAA devices are disclosed in U.S. Pat. No. 10,164,012, titled “Semiconductor Device and Manufacturing Method Thereof” and issued on Dec. 25, 2018, as well as in U.S. Pat. No. 10,361,278, titled “Method of Manufacturing a Semiconductor Device and a Semiconductor Device” and issued on Jul. 23, 2019, the disclosures of each which are hereby incorporated by reference in their respective entireties.

[0036] FIG. **17** illustrates a flowchart of a method **1000** for forming an IC device or a portion thereof from a workpiece, according to one or more aspects of the present disclosure. Method **1000** is merely an example and is not intended to limit the present disclosure to what is explicitly illustrated in method **1000**. Additional acts can be provided before, during and after the method **1000**, and some acts described can be replaced, eliminated, or moved around for additional embodiments of the methods. Not all acts are described herein in detail for reasons of simplicity. Method **1000** is described below in conjunction with fragmentary perspective and/or cross-sectional views of a workpiece, shown in FIGS. **2-16C**, at different stages of fabrication according to embodiments of method **1000**. For avoidance of doubt, throughout the figures, the X direction is perpendicular to the Y direction and the Z direction is perpendicular to both the X direction and the Y direction. It is noted that, because the workpiece may be fabricated into a semiconductor device, the workpiece may be referred to as the semiconductor device as the context requires.

[0037] FIGS. **2A** through **11** are perspective views and cross-sectional views of intermediate stages in the manufacturing of nano-FETs, in accordance with some embodiments. FIGS. **2A**, **3A**, **4A**, **5A**, **6A**, **7A**, **8A**, **9A**, and **10A** illustrate perspective views. FIGS. **2B**, **3B**, **4B**, **5B**, **6B**, **7B**, **8B**, **9B**, and **10B** illustrate reference cross-section B-B' (gate cut) illustrated in FIGS. **2A**, **3A**, and **4A**. FIGS. **4C**, **5C**, **6C**, **7C**, **8C**, **9C**, **10C** and **11** illustrate reference cross-section C-C' (channel/fin cut) illustrated in FIG. **4A**.

[0038] In FIG. **2A** and FIG. **2B**, a substrate **110** is provided. The substrate **110** may be a semiconductor substrate, such as a bulk semiconductor, or the like, which may be doped (e.g., with a p-type or an n-type dopant) or undoped. The semiconductor material of the substrate **110** may include silicon; germanium; a compound semiconductor including silicon carbide, gallium arsenide, gallium phosphide, indium phosphide, indium arsenide, and/or indium antimonide; an alloy semiconductor including silicon-germanium, gallium arsenide phosphide, aluminum indium arsenide, aluminum gallium arsenide, gallium indium arsenide, gallium indium phosphide, and/or

gallium arsenide phosphide; or combinations thereof. Other substrates, such as single-layer, multi-layered, or gradient substrates may be used.

[0039] Further in FIG. 2A and FIG. 2B, a multi-layer stack **25** or “lattice” is formed over the substrate **110** of alternating layers of first semiconductor layers **21A-21C** (collectively referred to as first semiconductor layers **21**) and second semiconductor layers **23A-23C** (collectively referred to as second semiconductor layers **23**). In some embodiments, the first semiconductor layers **21** may be formed of a first semiconductor material suitable for n-type nano-FETs, such as silicon, silicon carbide, or the like, and the second semiconductor layers **23** may be formed of a second semiconductor material suitable for p-type nano-FETs, such as silicon germanium or the like. Each of the layers of the multi-layer stack **25** may be epitaxially grown using a process such as chemical vapor deposition (CVD), atomic layer deposition (ALD), vapor phase epitaxy (VPE), molecular beam epitaxy (MBE), or the like.

[0040] Three layers of each of the first semiconductor layers **21** and the second semiconductor layers **23** are illustrated. In some embodiments, the multi-layer stack **25** may include one or two each or four or more each of the first semiconductor layers **21** and the second semiconductor layers **23**. Although the multi-layer stack **25** is illustrated as including a second semiconductor layer **23C** as the bottommost layer, in some embodiments, the bottommost layer of the multi-layer stack **25** may be a first semiconductor layer **21**.

[0041] Due to high etch selectivity between the first semiconductor materials and the second semiconductor materials, the second semiconductor layers **23** of the second semiconductor material may be removed without significantly removing the first semiconductor layers **21** of the first semiconductor material, thereby allowing the first semiconductor layers **21** to be patterned to form channel regions of nano-FETs. In some embodiments, the first semiconductor layers **21** are removed and the second semiconductor layers **23** are patterned to form channel regions. The high etch selectivity allows the first semiconductor layers **21** of the first semiconductor material to be removed without significantly removing the second semiconductor layers **23** of the second semiconductor material, thereby allowing the second semiconductor layers **23** to be patterned to form channel regions of nano-FETs.

[0042] In FIG. 3A and FIG. 3B, fins **32** are formed in the substrate **110** and nanostructures **22, 24** are formed in the multi-layer stack **25** corresponding to act **1100** of FIG. 17. In some embodiments, the nanostructures **22, 24** and the fins **32** may be formed by etching trenches in the multi-layer stack **25** and the substrate **110**. The etching may be any acceptable etch process, such as a reactive ion etch (RIE), neutral beam etch (NBE), the like, or a combination thereof. The etching may be anisotropic. First nanostructures **22A-22C** (also referred to as “channels” below) are formed from the first semiconductor layers **21**, and second nanostructures **24A-24C** are formed from the second semiconductor layers **23**. Distance **CD1** between adjacent fins **32** and nanostructures **22, 24** may be from about 18 nm to about 100 nm. A portion of the device **10** is illustrated in FIGS. 3A and 3B including two fins **32** for simplicity of illustration. The device **10** shown in FIGS. 1A-1F is shown including three fins **322-324**. The process **1000** illustrated in FIGS. 2A-16C may be extended to any number of fins, and is not limited to the two fins **32** shown in FIGS. 3A-16C nor to the three fins **322-324** shown in FIG. 1E.

[0043] The fins **32** and the nanostructures **22, 24** may be patterned by any suitable method. For example, one or more photolithography processes, including double-patterning or multi-patterning processes, may be used to form the fins **32** and the nanostructures **22, 24**. Generally, double-patterning or multi-patterning processes combine photolithography and self-aligned processes, allowing for pitches smaller than what is otherwise obtainable using a single, direct photolithography process. As an example of one multi-patterning process, a sacrificial layer may be formed over a substrate and patterned using a photolithography process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers may then be used to pattern the fins **32**.

[0044] FIGS. 3A and 3B illustrate the fins 32 having tapered sidewalls, such that a width of each of the fins 32 and/or the nanostructures 22, 24 continuously increases in a direction towards the substrate 110. In such embodiments, each of the nanostructures 22, 24 may have a different width and be trapezoidal in shape. In other embodiments, the sidewalls are substantially vertical (non-tapered), such that width of the fins 32 and the nanostructures 22, 24 is substantially similar, and each of the nanostructures 22, 24 is rectangular in shape.

[0045] In FIGS. 3A and 3B, isolation regions 36, which may be shallow trench isolation (STI) regions, are formed adjacent the fins 32. The isolation regions 36 may be formed by depositing an insulation material over the substrate 110, the fins 32, and nanostructures 22, 24, and between adjacent fins 32 and nanostructures 22, 24. The insulation material may be an oxide, such as silicon oxide, a nitride, the like, or a combination thereof, and may be formed by high-density plasma CVD (HDP-CVD), flowable CVD (FCVD), the like, or a combination thereof. In some embodiments, a liner (not separately illustrated) may first be formed along surfaces of the substrate 110, the fins 32, and the nanostructures 22, 24. Thereafter, a fill material, such as those discussed above may be formed over the liner.

[0046] The insulation material undergoes a removal process, such as a chemical mechanical polish (CMP), an etch-back process, combinations thereof, or the like, to remove excess insulation material over the nanostructures 22, 24. Top surfaces of the nanostructures 22, 24 may be exposed and level with the insulation material after the removal process is complete.

[0047] The insulation material is then recessed to form the isolation regions 36. After recessing, the nanostructures 22, 24 and upper portions of the fins 32 may protrude from between neighboring isolation regions 36. The isolation regions 36 may have top surfaces that are flat as illustrated, convex, concave, or a combination thereof. In some embodiments, the isolation regions 36 are recessed by an acceptable etching process, such as an oxide removal using, for example, dilute hydrofluoric acid (dHF), which is selective to the insulation material and leaves the fins 32 and the nanostructures 22, 24 substantially unaltered.

[0048] FIGS. 2A through 3B illustrate one embodiment (e.g., etch last) of forming the fins 32 and the nanostructures 22, 24. In some embodiments, the fins 32 and/or the nanostructures 22, 24 are epitaxially grown in trenches in a dielectric layer (e.g., etch first). The epitaxial structures may comprise the alternating semiconductor materials discussed above, such as the first semiconductor materials and the second semiconductor materials.

[0049] Further in FIG. 3A and FIG. 3B, appropriate wells (not separately illustrated) may be formed in the fins 32, the nanostructures 22, 24, and/or the isolation regions 36. Using masks, an n-type impurity implant may be performed in p-type regions of the substrate 110, and a p-type impurity implant may be performed in n-type regions of the substrate 110. Example n-type impurities may include phosphorus, arsenic, antimony, or the like. Example p-type impurities may include boron, boron fluoride, indium, or the like. An anneal may be performed after the implants to repair implant damage and to activate the p-type and/or n-type impurities. In some embodiments, in situ doping during epitaxial growth of the fins 32 and the nanostructures 22, 24 may obviate separate implantations, although in situ and implantation doping may be used together.

[0050] In FIGS. 4A-4C, dummy gate structures 40 are formed over the fins 32 and/or the nanostructures 22, 24. A dummy gate layer 45 is formed over the fins 32 and/or the nanostructures 22, 24. The dummy gate layer 45 may be made of materials that have a high etching selectivity versus the isolation regions 36. The dummy gate layer 45 may be a conductive, semiconductive, or non-conductive material and may be selected from a group including amorphous silicon, polycrystalline-silicon (polysilicon), poly-crystalline silicon-germanium (poly-SiGe), metallic nitrides, metallic silicides, metallic oxides, and metals. The dummy gate layer 45 may be deposited by physical vapor deposition (PVD), CVD, sputter deposition, or other techniques for depositing the selected material. A mask layer 47 is formed over the dummy gate layer 45, and may include, for example, silicon nitride, silicon oxynitride, or the like. In some embodiments, a gate dielectric

layer (not illustrated for simplicity) is formed before the dummy gate layer **45** between the dummy gate layer **45** and the fins **32** and/or the nanostructures **22**, **24**.

[0051] A spacer layer **41** is formed over sidewalls of the mask layer **47** and the dummy gate layer **45**. The spacer layer **41** is made of an insulating material, such as silicon nitride, silicon oxide, silicon carbo-nitride, silicon oxynitride, silicon oxy carbo-nitride, or the like, and may have a single-layer structure or a multi-layer structure including a plurality of dielectric layers, in accordance with some embodiments. The spacer layer **41** may be formed by depositing a spacer material layer (not shown) over the mask layer **47** and the dummy gate layer **45**. Portions of the spacer material layer between dummy gate structures **40** are removed using an anisotropic etching process, in accordance with some embodiments.

[0052] FIGS. **4A-4C** illustrate one process for forming the spacer layer **41**. In some embodiments, the spacer layer **41** is formed alternately or additionally after removal of the dummy gate layer **45**. In such embodiments, the dummy gate layer **45** is removed, leaving an opening, and the spacer layer **41** may be formed by conformally coating material of the spacer layer **41** along sidewalls of the opening. The conformally coated material may then be removed from the bottom of the opening corresponding to the top surface of the uppermost channel, e.g., the channel **22A**, prior to forming an active gate, such as any of the gate structures **200A-200E**.

[0053] In FIGS. **5A-5C**, an etching process is performed to etch the portions of protruding fins **32** and/or nanostructures **22**, **24** that are not covered by dummy gate structures **40**, resulting in the structure shown. The recessing may be anisotropic, such that the portions of fins **32** directly underlying dummy gate structures **40** and the spacer layer **41** are protected, and are not etched. The top surfaces of the recessed fins **32** may be substantially coplanar with the top surfaces of the isolation regions **36** as shown, in accordance with some embodiments. The top surfaces of the recessed fins **32** may be lower than the top surfaces of the isolation regions **36**, in accordance with some other embodiments. FIG. **5C** shows two vertical stacks of nanostructures **22**, **24** following the etching process for simplicity. FIGS. **1B**, **1D**, **1F** show five vertical stacks of nanostructures **22** (nanostructures **24** are replaced by gate structures **200A-200E**). In general, the etching process may be used to form any number of vertical stacks of nanostructures **22**, **24** over the fins **32** (or over the fins **322-324**).

[0054] FIGS. **6A-6C** and **7A-7C** illustrate formation of inner spacers **74**. A selective etching process is performed to recess end portions of the nanostructures **24** exposed by openings in the spacer layer **41** without substantially attacking the nanostructures **22**. After the selective etching process, recesses **64** are formed in the nanostructures **24** at locations where the removed end portions used to be. The resulting structure is shown in FIGS. **6A-6C**.

[0055] Next, an inner spacer layer is formed to fill the recesses **64** in the nanostructures **22** formed by the previous selective etching process. The inner spacer layer may be a suitable dielectric material, such as silicon carbon nitride (SiCN), silicon oxycarbonitride (SiOCN), or the like, formed by a suitable deposition method such as PVD, CVD, ALD, or the like. An etching process, such as an anisotropic etching process, is performed to remove portions of the inner spacer layers disposed outside the recesses in the nanostructures **24**. The remaining portions of the inner spacer layers (e.g., portions disposed inside the recesses **64** in the nanostructures **24**) form the inner spacers **74**. The resulting structure is shown in FIGS. **7A-7C**.

[0056] FIGS. **8A-8C** illustrate formation of source/drain regions **82** corresponding to act **1200** of FIG. **17**. In the illustrated embodiment, the source/drain regions **82** are epitaxially grown from epitaxial material(s). In some embodiments, the source/drain regions **82** exert stress in the respective channels **22A-22C**, thereby improving performance. The source/drain regions **82** are formed such that each dummy gate structure **40** is disposed between respective neighboring pairs of the source/drain regions **82**. In some embodiments, the spacer layer **41** separates the source/drain regions **82** from the dummy gate layer **45** by an appropriate lateral distance to prevent electrical bridging to subsequently formed gates of the resulting device.

[0057] The source/drain regions **82** may include any acceptable material, such as appropriate for n-type or p-type devices. For n-type devices, the source/drain regions **82** include materials exerting a tensile strain in the channel regions, such as silicon, SiC, SiCP, SiP, or the like, in some embodiments. When p-type devices are formed, the source/drain regions **82** include materials exerting a compressive strain in the channel regions, such as SiGe, SiGeB, Ge, GeSn, or the like, in accordance with certain embodiments. The source/drain regions **82** may have surfaces raised from respective surfaces of the fins and may have facets. Neighboring source/drain regions **82** may merge in some embodiments to form a singular source/drain region **82** adjacent two neighboring fins **32**.

[0058] The source/drain regions **82** may be implanted with dopants followed by an anneal. The source/drain regions may have an impurity concentration of between about 10^{19} cm⁻³ and about 10^{21} cm⁻³. N-type and/or p-type impurities for source/drain regions **82** may be any of the impurities previously discussed. In some embodiments, the source/drain regions **82** are in situ doped during growth. A contact etch stop layer (CESL) **131** and interlayer dielectric (ILD) **130**, shown in FIG. **11**, may then be formed covering the dummy gate structures **40** and the source/drain regions **82**. The CESL **131** and ILD **130** are omitted from FIGS. **8A-10C** for simplicity of illustration.

[0059] FIG. **9A**, FIG. **9B**, and FIG. **9C** illustrate release of fin channels **22A-22C** by removal of the nanostructures **24A-24C**, the mask layer **47**, and the dummy gate layer **45**, which corresponds to act **1300** of FIG. **17**. A planarization process, such as a CMP, is performed to level the top surfaces of the dummy gate layer **45** and gate spacer layer **41**. The planarization process may also remove the mask layer **47** (see FIG. **8A**) on the dummy gate layer **45**, and portions of the gate spacer layer **41** along sidewalls of the mask layer **47**. Accordingly, the top surfaces of the dummy gate layer **45** are exposed.

[0060] Next, the dummy gate layer **45** is removed in an etching process, so that recesses **92** are formed. In some embodiments, the dummy gate layer **45** is removed by an anisotropic dry etch process. For example, the etching process may include a dry etch process using reaction gas(es) that selectively etch the dummy gate layer **45** without etching the spacer layer **41**. The dummy gate dielectric, when present, may be used as an etch stop layer when the dummy gate layer **45** is etched. The dummy gate dielectric may then be removed after the removal of the dummy gate layer **45**.

[0061] The nanostructures **24** are removed to release the nanostructures **22**. After the nanostructures **24** are removed, the nanostructures **22** form a plurality of nanosheets that extend horizontally (e.g., parallel to a major upper surface of the substrate **110**). The nanosheets may be collectively referred to as the channels **22** of the GAA devices **20A-20E** formed.

[0062] In some embodiments, the nanostructures **24** are removed by a selective etching process using an etchant that is selective to the material of the nanostructures **24**, such that the nanostructures **24** are removed without substantially attacking the nanostructures **22**. In some embodiments, the etching process is an isotropic etching process using an etching gas, and optionally, a carrier gas, where the etching gas comprises F₂ and HF, and the carrier gas may be an inert gas such as Ar, He, N₂, combinations thereof, or the like.

[0063] In some embodiments, the nanostructures **24** are removed and the nanostructures **22** are patterned to form channel regions of both PFETs and NFETs. However, in some embodiments the nanostructures **24** may be removed and the nanostructures **22** may be patterned to form channel regions of NFETs, and nanostructures **22** may be removed and the nanostructures **24** may be patterned to form channel regions of PFETs. In some embodiments, the nanostructures **22** may be removed and the nanostructures **24** may be patterned to form channel regions of NFETs, and the nanostructures **24** may be removed and the nanostructures **22** may be patterned to form channel regions of PFETs. In some embodiments, the nanostructures **22** may be removed and the nanostructures **24** may be patterned to form channel regions of both PFETs and NFETs.

[0064] In some embodiments, the nanosheets **22** are reshaped (e.g. thinned) by a further etching process to improve gate fill window. The reshaping may be performed by an isotropic etching process selective to the nanosheets **22**. After reshaping, the nanosheets **22** may exhibit the dog bone shape in which middle portions of the nanosheets **22** are thinner than peripheral portions of the nanosheets **22** along the X direction.

[0065] In FIGS. **10A-10C**, replacement gates **200** are formed, corresponding to act **1400** of FIG. **17**. FIG. **12** is a detailed view of the region **170** of FIG. **10B** corresponding to a portion of the gate structure **200**. The gate structure **200** generally includes the interfacial layer (IL, or “first IL” below) **210**, at least one gate dielectric layer **600**, the work function metal layer **900**, and the gate fill layer **290**. In some embodiments, each replacement gate **200** further includes at least one of a second interfacial layer **240** or a second work function layer **700**.

[0066] With reference to FIG. **12**, in some embodiments, the first IL **210** includes an oxide of the semiconductor material of the substrate **110**, e.g. silicon oxide. In other embodiments, the first IL **210** may include another suitable type of dielectric material. The first IL **210** has a thickness in a range between about 5 angstroms and about 50 angstroms.

[0067] Still referring to FIG. **12**, the gate dielectric layer **600** is formed over the first IL **210**. In some embodiments, an atomic layer deposition (ALD) process is used to form the gate dielectric layer **600** to control thickness of the deposited gate dielectric layer **600** with precision. In some embodiments, the ALD process is performed using between about 40 and 80 deposition cycles, at a temperature range between about 200 degrees Celsius and about 300 degrees Celsius. In some embodiments, the ALD process uses HfCl_4 and/or H_2O as precursors. Such an ALD process may form the first gate dielectric layer **600** to have a thickness in a range between about 10 angstroms and about 100 angstroms.

[0068] In some embodiments, the gate dielectric layer **600** includes a high-k dielectric material, which may refer to dielectric materials having a high dielectric constant that is greater than a dielectric constant of silicon oxide ($k \approx 3.9$). Exemplary high-k dielectric materials include $\text{HfO}_{2.2}$, HfSiO , HfSiON , HfTaO , HfTiO , HfZrO , $\text{ZrO}_{2.2}$, $\text{Ta}_{2.5}\text{O}_{2.5}$, or combinations thereof. In other embodiments, the gate dielectric layer **600** may include a non-high-k dielectric material such as silicon oxide. In some embodiments, the gate dielectric layer **600** includes more than one high-k dielectric layer, of which at least one includes dopants, such as lanthanum, magnesium, yttrium, or the like, which may be driven in by an annealing process to modify threshold voltage of the GAA devices **20A-20E**.

[0069] With further reference to FIG. **12**, the second IL **240** is formed on the gate dielectric layer **600**, and the second work function layer **700** is formed on the second IL **240**. The second IL **240** promotes better metal gate adhesion on the gate dielectric layer **600**. In many embodiments, the second IL **240** further provides improved thermal stability for the gate structure **200**, and serves to limit diffusion of metallic impurity from the work function metal layer **900** and/or the work function barrier layer **700** into the gate dielectric layer **600**. In some embodiments, formation of the second IL **240** is accomplished by first depositing a high-k capping layer (not illustrated for simplicity) on the gate dielectric layer **600**. The high-k capping layer comprises one or more of the following: HfSiON , HfTaO , HfTiO , HfAlON , HfZrO , or other suitable materials, in various embodiments. In a specific embodiment, the high-k capping layer comprises titanium silicon nitride (TiSiN). In some embodiments, the high-k capping layer is deposited by an ALD using about 40 to about 100 cycles at a temperature of about 400 degrees C. to about 450 degrees C. A thermal anneal is then performed to form the second IL **240**, which may be or comprise TiSiNO , in some embodiments. Following formation of the second IL **240** by thermal anneal, an atomic layer etch (ALE) with artificial intelligence (AI) control may be performed in cycles to remove the high-k capping layer while substantially not removing the second IL **240**. Each cycle may include a first pulse of WCl_5 , followed by an Ar purge, followed by a second pulse of O_2 , followed by another Ar purge. The high-k capping layer is removed to increase gate fill window for

further multiple threshold voltage tuning by metal gate patterning.

[0070] Further in FIG. 12, after forming the second IL **240** and removing the high-k capping layer, the work function barrier layer **700** is optionally formed on the gate structure **200**, in accordance with some embodiments. The work function barrier layer **700** is or comprises a metal nitride, such as TiN, WN, MON, TaN, or the like. In a specific embodiment, the work function barrier layer **700** is TiN. The work function barrier layer **700** may have thickness ranging from about 5 Å to about 20 Å. Inclusion of the work function barrier layer **700** provides additional threshold voltage tuning flexibility. In general, the work function barrier layer **700** increases the threshold voltage for NFET transistor devices, and decreases the threshold voltage (magnitude) for PFET transistor devices.

[0071] The work function metal layer **900**, which may include at least one of an N-type work function metal layer, an in-situ capping layer, or an oxygen blocking layer, is formed on the work function barrier layer **700**, in some embodiments. The N-type work function metal layer is or comprises an N-type metal material, such as TiAlC, TiAl, TaAlC, TaAl, or the like. The N-type work function metal layer may be formed by one or more deposition methods, such as CVD, PVD, ALD, plating, and/or other suitable methods, and has a thickness between about 10 Å and 20 Å. The in-situ capping layer is formed on the N-type work function metal layer. In some embodiments, the in-situ capping layer is or comprises TiN, TiSiN, TaN, or another suitable material, and has a thickness between about 10 Å and 20 Å. The oxygen blocking layer is formed on the in-situ capping layer to prevent oxygen diffusion into the N-type work function metal layer, which would cause an undesirable shift in the threshold voltage. The oxygen blocking layer is formed of a dielectric material that can stop oxygen from penetrating to the N-type work function metal layer, and may protect the N-type work function metal layer from further oxidation. The oxygen blocking layer may include an oxide of silicon, germanium, SiGe, or another suitable material. In some embodiments, the oxygen blocking layer is formed using ALD and has a thickness between about 10 Å and about 20 Å.

[0072] FIG. 12 further illustrates the metal fill layer **290**. In some embodiments, a glue layer (not separately illustrated) is formed between the oxygen blocking layer of the work function metal layer and the metal fill layer **290**. The glue layer may promote and/or enhance the adhesion between the metal fill layer **290** and the work function metal layer **900**. In some embodiments, the glue layer may be formed of a metal nitride, such as TiN, TaN, MON, WN, or another suitable material, using ALD. In some embodiments, thickness of the glue layer is between about 10 Å and about 25 Å. The metal fill layer **290** may be formed on the glue layer, and may include a conductive material such as tungsten, cobalt, ruthenium, iridium, molybdenum, copper, aluminum, or combinations thereof. In some embodiments, the metal fill layer **290** may be deposited using methods such as CVD, PVD, plating, and/or other suitable processes. In some embodiments, a seam **510**, which may be an air gap, is formed in the metal fill layer **290** vertically between the channels **22A**, **22B**. In some embodiments, the metal fill layer **290** is conformally deposited on the work function metal layer **900**. The seam **510** may form due to sidewall deposited film merging during the conformal deposition. In some embodiments, the seam **510** is not present between the neighboring channels **22A**, **22B**.

[0073] FIGS. 13A-16C illustrate formation of the source/drain contact isolation structures **150** in accordance with various embodiments. The source/drain contact isolation structures **150** may be formed prior to formation of the source/drain contacts **120** (see FIG. 1E, for example).

[0074] FIGS. 13C, 14C, 15C, 16C are cross-sectional top views of the device **10** taken along the line C-C shown in FIGS. 13A, 13B, 14A, 14B, 15B, 15C, 16C, 16D. FIGS. 13A, 14A, 15A, 16A are cross-sectional side views of the device **10** taken along the line A-A shown in FIGS. 13C, 14C, 15C, 16C. FIGS. 13B, 14B, 15B, 16B are cross-sectional side views of the device **10** taken along the line B-B shown in FIGS. 13C, 14C, 15C, 16C.

[0075] In FIGS. 13A-13C, following formation of the gate structures **200A-200E**, overlying the gate dielectric layer **600** and the gate fill layer **290** are a conductive layer **204**, an optional hard

dielectric layer (not shown), and a capping layer **295**. The conductive layer **204** reduces contact resistance between a gate contact (not shown) and the gate fill layer **290**. In some embodiments, the conductive layer **204** comprises a metal, such as tungsten, aluminum, copper, ruthenium, or the like.

[0076] The capping layer **295**, also referred to as a “self-aligned capping” (SAC) layer, may provide protection to the underlying gate structure **200A-200E**, and may also act as a CMP stop layer when planarizing the source/drain contacts **120** following formation thereof. The capping layer **295** may be a dielectric layer including a dielectric material, such as SiO₂, SiN, SiCN, SiC, SiOC, SiOCN, HfO₂, ZrO₂, ZrAlO_x, HfAlO_x, HfSiO_x, Al₂O₃, BN, or other suitable dielectric material. Between the capping layer **295** and the conductive layer **204** is the optional hard dielectric layer. The hard dielectric layer may prevent current leakage following one or more etching operations, which may be performed to form gate contacts, source/drain contacts **120**, isolation structures (e.g., source/drain contact isolation structures **150**), or the like. In some embodiments, the hard dielectric layer is or comprises a dielectric material that is harder than, for example, the capping layer **295**, such as aluminum oxide, or other suitable dielectric material. The hard dielectric layer may also be between the capping layer **295** and the spacer layer **41**. In some embodiments, over gate structures **200D**, **200E** and channels **22A4-22B5**, the capping layer **295** may be split by a support structure **297**, as shown. In some embodiments, width (X direction) of the capping layer **295** is in a range of about 8 nm to about 40 nm.

[0077] In FIGS. **14A-14C**, openings **775** are formed, corresponding to act **1500** of FIG. **17**. The source/drain contact isolation structures **150** are formed in the openings **775** in subsequent operations. The openings **775** may be formed by one or more masking operations followed by one or more etching operations. In some embodiments, a first masking layer **1410** is formed over the top surface of the device **10**, followed by a second masking layer **1420** formed over the first masking layer **1410**. The first and second masking layers **1410**, **1420** may be any appropriate masking layer, such one or more of a photoresist layer, a hard mask layer, an anti-reflective coating layer, or the like. Following formation of the first and second masking layers **1410**, **1420**, the openings **775** are formed to the upper surface of the ILD **130** and the CESL **131** by patterning the first and second masking layers **1410**, **1420**. The patterning may include exposing the second masking layer **1420** to light, such as deep ultraviolet (DUV) or extreme ultraviolet (EUV) light, then removing exposed or unexposed portions of the second masking layer **1420**, for example, by a rinsing step. Following patterning of the second masking layer **1420**, the openings **775** may be extended into the first masking layer **1420** by, for example, an etching operation. Following patterning of the first masking layer **1420**, the openings **775** may expose the capping layer **295**, the ILD **130** and the CESL **131**.

[0078] Following exposing the capping layer **295**, the ILD **130** and the CESL **131**, the ILD **130** may be removed by one or more removal operations. The removal operations may include an etching operation that is selective to the ILD **130** compared to the CESL **131** and the capping layer **295**.

[0079] Following removal of the ILD **130**, the CESL **131** is exposed in the openings **775**. The openings **775** are expanded by trimming (e.g., tapered) the CESL **131** by one or more etching operations that is selective to the CESL **131** compared to the capping layer **295**, the source/drain region **82**, and the spacer layer **41**, corresponding to act **1600** of FIG. **17**. In some embodiments, the CESL **131** is removed completely where exposed by the openings **775**, as shown in FIG. **14B**. FIG. **16B** shows a partially trimmed CESL **131** in accordance with various embodiments. In some embodiments, the trimming forms a tapered profile of reduced thickness in the CESL **131**, as shown by the CESL **131CMD** in FIG. **1F** and FIG. **16B**. In some embodiments, the trimming reduces thickness of the CESL **131** in the openings **775** by at least about 50%, such as by about 60% or as much as 100% (complete removal).

[0080] Prior to etching of the CESL **131**, the portions of the openings **775** below the upper surface

of the capping layer **295** may each have an aspect ratio (height/width) that is greater than about 7, which may lead to formation of voids in subsequent deposition of the source/drain contact isolation structures **150**. Following etching of the CESL **131**, the portions of the openings **775** are expanded, and may each have an aspect ratio less than about 5, which prevents the formation of the voids in the source/drain contact isolation structures **150**. The openings **775** land on (e.g., expose) the hybrid fin **94**, such as the fill layer **93** of the hybrid fin **94**.

[0081] In FIGS. **15A-15C**, the first and second masking layers **1410**, **1420** are removed, for example, by ashing, CMP or another suitable method, and the source/drain contact isolation structures **150** are formed by filling in the openings **775**, corresponding to act **1700** of FIG. **17**. In some embodiments, the source/drain contact isolation structures **150** are formed by depositing a dielectric material in the openings **775** by, for example, CVD, ALD, or another suitable process. The dielectric material may be or include one or more of SIN, SiCN, SiC, SiOC, SiOCN, HfO.sub.2, ZrO.sub.2, ZrAlO.sub.x, HfAlO.sub.x, HfSiO.sub.x, Al.sub.2O.sub.3. Following filling of the openings **775** with the dielectric material, the source/drain contact isolation structures **150** have aspect ratio (height/width) less than about 5. In some embodiments, height of the source/drain contact isolation structures **150** may be in a range of about 15 nm to about 40 nm. In some embodiments, width (X direction) of the source/drain contact isolation structures **150** may be in a range of about 8 nm to about 30 nm. In some embodiments, the aspect ratio is in a range of about 0.5 to about 5. The reduced aspect ratio compared to when the CESL **131** is present and untrimmed (e.g., about 5 nm in thickness) provides benefits of increased production yield, better filling of the dielectric material (e.g., substantially no voids present) and an expanded potential pool of dielectric materials available for use in isolating the source/drain contacts **120**.

[0082] In FIGS. **16A-16C**, and with reference to FIG. **16B**, in some embodiments, the CESL **131** in the openings **775** is partially removed, e.g., trimmed and tapered, which forms the CESL **131CMD** shown. The trimming may include stopping the etching operation used to remove the CESL **131** prior to full removal of the CESL **131**. Following the partial removal of the CESL **131** to form the CESL **131CMD**, the width of the source/drain contact isolation structures **150** may vary (e.g., decrease) in a direction toward the substrate **110**. In some embodiments, greatest thickness of the CESL **131CMD** lining the source/drain contact isolation structures **150** may be in a range of about 0.5 nm to about 2 nm, and least thickness of the CESL **131CMD** approaches 0 nm due to tapering. In some embodiments, greatest width of the source/drain contact isolation structure **150** may be in a range of about 8 nm to about 30 nm, and least width of the source/drain contact isolation structure **150** may be in a range of about 6 nm to about 29.5 nm. Due to tapering of the CESL **131CMD**, aspect ratio of the source/drain contact isolation structures **150** may be calculated differently based on trapezoidal shape of the source/drain contact isolation structures **150**. Assuming a smooth taper of the CESL **131CMD**, the aspect ratio may be equal to height of the source/drain contact isolation structure **150** over average width of the source/drain contact isolation structure **150**. In some embodiments, the aspect ratio of the trapezoidal source/drain contact isolation structure **150** may be in a range of about 0.5 to about 6. The reduced aspect ratio compared to when the CESL **131** is present and untrimmed (e.g., about 5 nm in thickness) provides benefits of increased production yield, better filling of the dielectric material (e.g., substantially no voids present) and an expanded potential pool of dielectric materials available for use in isolating the source/drain contacts **120**. In addition to these benefits, presence of the CESL **131CMD** may provide additional structural and/or electrical protection to the spacer layers **41** and the gate structures **200A-200E**. Presence or removal of the CESL **131CMD** may be weighed based on desire for better filling performance (thinner or no CESL **131CMD**) or better protection (thicker CESL **131CMD**).

[0083] Additional processing may be performed to finish fabrication of the GAA devices **20A-20E**. For example, gate contacts **184** (see FIG. **1B**) may be formed to electrically couple to the gate structures **200A-200E**, such as the gate structure **200C**. An interconnect structure may then be formed over the source/drain contacts **120** and the gate contacts **184**. The interconnect structure

may include a plurality of dielectric layers surrounding metallic features, including conductive traces and conductive vias, which form electrical connection between devices on the substrate **110**, such as the GAA devices **20A-20E**, as well as to IC devices external to the IC device **10**. In some embodiments, second capping layers (not shown) are present over the source/drain contacts **120**. Configurations in which only the capping layers **295** over the gate structures **200A-200E** are present (e.g., no second capping layers present over the source/drain contacts **120**) may be considered “single SAC” structures, and configurations in which the capping layers **295** and the second capping layers are both present may be considered “double SAC” structures. The source/drain contact isolation structures **150** with reduced aspect ratio are compatible with both single SAC and double SAC configurations.

[0084] Embodiments may provide advantages. By removing or thinning the CESL **131** prior to filling in the source/drain contact isolation structures **150**, the aspect ratio of the openings **775** in which the source/drain contact isolation structures **150** are filled is reduced, which reduces the likelihood of formation of voids. As such, production yield may be increased, filling performance is improved and the potential material pool for the source/drain contact isolation structures **150** is expanded.

[0085] In accordance with at least one embodiment, a device includes: a substrate; a gate structure wrapping around at least one vertical stack of nanostructure channels; a source/drain region abutting the gate structure; a source/drain contact over the source/drain region; an etch stop layer laterally between the source/drain contact and the gate structure and having a first sidewall in contact with the source/drain contact, and a second sidewall opposite the first sidewall; and a source/drain contact isolation structure embedded in the source/drain contact and having a third sidewall substantially coplanar with the second sidewall of the etch stop layer.

[0086] In accordance with at least one embodiment, a device includes: a substrate; a first source/drain region in and on the substrate; a second source/drain region laterally separated from the first source/drain region; a gate structure between the first and second source/drain regions; a source/drain contact over the first source/drain region, and having a first sidewall and a second sidewall opposite the first sidewall; a source/drain contact isolation structure over the second source/drain region, and having a third sidewall and a fourth sidewall opposite the third sidewall; first liner layers lining the first and second sidewalls; and second liner layers lining the third and fourth sidewalls, the second liner layers thinner than the first liner layers.

[0087] In accordance with at least one embodiment, a method includes: forming a first source/drain region and a second source/drain region on and in a substrate, the first source/drain region laterally separated from the second source/drain region; forming a first liner layer over the first source/drain region and a second liner layer over the second source/drain region; forming a first fill layer over the first liner layer and a second fill layer over the second liner layer; forming an opening in the first fill layer; expanding the opening by trimming the first liner layer; and forming an isolation structure by filling the expanded opening with a dielectric material.

[0088] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

- 1.** A device, comprising: a substrate; a gate structure wrapping around one or more semiconductor channels; a source/drain region abutting the gate structure; a source/drain contact over the source/drain region; an etch stop layer laterally between the source/drain contact and the gate structure; and a source/drain contact isolation structure formed within the source/drain contact, wherein the aspect ratio (height/width) of the source/drain contact isolation structure is less than the aspect ratio (height/width) of the source/drain contact.
- 2.** The device of claim 1, further comprising a capping layer overlying the gate structure, wherein the capping layer has sidewall in contact with a sidewall of the etch stop layer and a sidewall of the source/drain contact isolation structure.
- 3.** The device of claim 2, further comprising a first gate spacer layer abutting the gate structure.
- 4.** The device of claim 1, wherein the source/drain contact isolation structure has a sidewall in contact with the source/drain contact and the etch stop layer.
- 5.** The device of claim 1, wherein: the source/drain contact is in contact with the source/drain region; and the source/drain contact isolation structure is in contact with a hybrid fin abutting the source/drain region.
- 6.** The device of claim 5, wherein: the source/drain contact is in contact with a silicide region of the source/drain region; and the source/drain contact isolation structure is in contact with an epitaxial region of the source/drain region.
- 7.** A device comprising: a substrate; a first source/drain region over the substrate; a second source/drain region laterally separated from the first source/drain region; a gate structure between the first and second source/drain regions; a source/drain contact over the first source/drain region; a source/drain contact isolation structure over the second source/drain region, the aspect ratio (height/width) of the source/drain contact isolation structure is less than the aspect ratio (height/width) of the source/drain contact; first liner layers lining a wall of the source/drain contact; and second liner layers lining a wall of the source/drain contact isolation structure, the second liner layers thinner than the first liner layers.
- 8.** The device of claim 7, wherein: the first liner layers have thickness in a range of about 1 nm to about 5 nm.
- 9.** The device of claim 7, wherein the second liner layers have a tapered profile that is wider proximal the substrate.
- 10.** The device of claim 7, wherein the aspect ratio of the source/drain contact isolation structure is less than 5.
- 11.** The device of claim 7, wherein thickness of the second liner layers is in a range from about 0.5 nm to about 2 nm.
- 12.** The device of claim 7, wherein extension portions of the second liner layers extend beyond the source/drain contact isolation structure, and thickness of the extension portions is substantially the same as thickness of the first liner layers.
- 13.** The device of claim 7, wherein the source/drain contact isolation structure has a tapered profile that is narrower proximal the substrate.
- 14.** A method, comprising: forming source/drain regions in and on a substrate, the source/drain regions being laterally separated from each other; forming liner and fill layers over the source/drain regions; forming at least one opening in at least one of the fill layers between adjacent source/drain regions; expanding the opening; forming an isolation structure by filling the expanded opening with a dielectric material; forming source/drain contact openings by removing at least a portion of the fill layers after forming the isolation structure; and forming at least three source/drain contacts by filling the source/drain contact openings; wherein the aspect ratio (height/width) of the isolation structure is less than the aspect ratio (height/width) of the source/drain contacts.
- 15.** The method of claim 14, wherein forming the isolation structure includes forming the isolation structure having a tapered profile that is narrower proximal the substrate.

- 16.** The method of claim 14, wherein expanding the opening comprises forming a tapered profile with reduced thickness in the liner layer by an etching process.
- 17.** The method of claim 14, wherein expanding the opening comprises trimming the liner layer, and further includes reducing thickness of the liner layer by at least 50%.
- 18.** The method of claim 17, wherein trimming the liner layer includes completely removing the liner layer in the opening.
- 19.** The method of claim 14 wherein the aspect ratio of the isolation structure is less than 5.
- 20.** The method of claim 19, further comprising: forming first self-aligned capping layers over a gate structure between adjacent source/drain regions; and forming second self-aligned capping layers over the source/drain contacts.
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